

Homework for week 2

- 1) Calculate the Fourier series of the 2π -periodic function given by

$$f(x) = x^2 \quad -\pi \leq x < \pi$$

Using the Parseval's identity, prove that

$$\sum_{n=1}^{\infty} \frac{1}{n^4} = \frac{\pi^4}{90}$$

- 2) Kreyszig Problem 11.3.17

Solve the RLC circuit using the same method as we used for the forced oscillations of a point mass.

- 3) Calculate the Fourier sine and cosine transforms of the functions given in the following exercises in the Kreyszig textbook.
(we covered some of these in the exercise class):

11.9 problems 3-7, 9-11